MILITARY SPECIFICATION

PROJECTILE ATOMIC EXPLOSIVE 8 INCH, M422 SHIPPING ASSEMBLY

1. SCOPE

1.1 Scope. This specification covers special requirements, quality assurance provisions, and the preparation for delivery criteria for one type of projectile atomic explosive 8 Inch, M422 shipping assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited)
QQ-Z-325 - Zinc Plating (Electrodeposited)
TT-C-490 - Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings

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MIL-A-8625 - Anodic Coatings for Aluminum and Aluminimum Alloys
MIL-B-20292 - Blanks, Rotating Band for Projectile
MIL-B-20295 - Blanks, Centrifugal Cast, Rotating Band
MIL-C-50565 - Container, Shipping and Storage, Atomic Projectile: M500
MIL-D-48016 - Dashpot Assembly, Atomic Weapon
MIL-D-50837 - Disc, Solid Plain
MIL-G-48204 - Grommet Assembly
MIL-P-48194 - Propellant Charge Nuclear Component

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MIL-P-50835 - Plug, Machine Thread
MIL-P-50891 - Beryllium Parts

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering for Ammunition

DRAWINGS

ORDNANCE CORPS

8786586 - Support Assembly
8786587 - Projectile, Atomic Explosive 8 Inch H, M422 Shipping Assembly
8787101 - Plug, Nose Atomic Projectile
8787102 - Ogive, Atomic Projectile
8787104 - Plug, Machine Thread
8787105 - Body Section, Atomic Projectile, Front
8787107 - Cylinder, Sintered
8787108 - Plug, Closing
8787112 - Disc, Solid Plain
8787113 - Nut, Plain Round
8787135 - Propellant Charge Nuclear Component
8787141 - Support Disc, Nuclear Propellant Charge
8787142 - Base, Atomic Projectile
8787144 - Base Plug, Atomic Projectile
8787145 - Spring, Helical Compression
8787146 - Cap, Spring
8787147 - Base Jacket Ring Assembly
8787149 - Base Jacket Assembly
8787150 - Ring
8787153 - Jacket Base Assembly
8787154 - Cone, Disc
8787163 - Body Section, Projectile, Rear
8787164 - Body Assembly
8787166 - Gun Tube Atomic Weapon
8787167 - Base, "O" Ring
8787239 - Stop, Nuclear Target
8787240 - Bushing, Machine Thread
8787241 - Container, Metal, Amo, for Projectile, 8 Inch, M422 Shipping Assembly

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8787244 - Container, Metal Shipping for Accessory Parts for Projectile 8 Inch, M422
8794227 - Dashpot Assembly, Atomic Weapon
8794333 - Spacer, Sleeve
8794529 - Jacket, Base
8795639 - Gasket
8795680 - Grommet Assembly
8804300 - Pin, Shoulder Headless
8804302 - Screw, Machine
8822568 - Base Plug Assembly
ET-8786587 - Equipment List

PUBLICATIONS

INDUSTRIAL

ASTM-E8-66 - Methods of Tension Testing of Metallic Materials

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The projectile shall comply with all requirements specified on the drawing (dwg) 8786587, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Tool manufacturing control (T.M.C.). When T.M.C. is specified herein, five consecutive parts which comply with drawing dimensions will constitute proper initial tool control acceptance. During normal production, five parts shall be selected at the start and at the end of each day's production (for a minimum of 10 parts) and examined as specified in the applicable paragraph in 4.3.2.1. In addition, five consecutive parts shall be examined as specified in the applicable paragraph in 4.3.2.1 whenever the manufacturing tool requires maintenance. Failure of any part to comply with any of the examinations specified in 4.3.2.1 shall reject the lot of production under consideration.

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3.5 Rotating band and obturating band.

3.5.1 Blanks. The rotating band and obturating band blanks shall be in the form of a solid ring, annealed if necessary, to comply with Specification MIL-B-20292 or MIL-B-20295. The band blanks shall be carefully pressed into the band seat so as to completely fill it and fit tightly.

3.5.2 Physical properties, rotating band and obturation band. After removal from the body, a 2-inch gage length test coupon shall show the following physical properties: Tensile 30,000 pounds per square inch (p.s.i.), elongation 25 percent minimum, when tested as specified in 4.4.1.

3.5.3 Seating

3.5.3.1 Rotating band. After seating of the band, the diametral clearance between the band and the seat shall not exceed .007 inch on stations 1 and 5, and shall not exceed .020 inch on stations 2, 3, and 4, when tested and completed as specified in 4.4.1. After banding, the height of the waves shall not be reduced more than 25 percent of the original height. (For inspection purposes the following stations have been established: Stations 1 and 5 are those points 1/2 inch toward the center of the band, as measured from the front and rear of the band respectively; Stations 2 and 4 are those points 3/4 inch toward the center of the band as measured from the front and rear of the band respectively and Station 3 is at the center of the band.)

3.5.3.2 Obturating band. After seating of the band, the diametral clearance between the band and the seat shall not exceed .007 inch when tested as specified in 4.4.1. For inspection purposes the leading and trailing edges of the band shall be measured.

3.6 Cup spring distortion. The springs shall not break when pulled to the minimum point of permanent distortion. Testing shall be as specified in 4.4.4.

3.7 Seating of rubber ring (see dwg. 8787147). The rubber ring shall completely fill the dovetail groove in the base jacket as determined by visual examination in accordance with paragraph 4.4.5.

3.8 Air pressure test. Each container shall withstand a pressure of 3 p.s.i. for 15 seconds without leakage when tested as specified in 4.4.6.

3.9 Serialization. The serial number of each assembly shall be duplicated on the exterior of the ammunition containers for accessory parts and on the exterior of the ammunition container assembly for the projectile. These serial numbers shall be assigned to each assembly by the procurement agency. The serial number shall be in addition to the normal lot numbers but in no event shall there by any possibility of duplication nor any indication or order or size of production in either the lot designation, the serial number, or any other number.

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3.10 Workmanship.

3.10.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.10.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.10.3 Burrs. Parts shall be free from burrs.

3.10.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosions, or any embedded foreign material.

3.10.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted (see 6.4). When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8786587) and 1 sub-assembly see dwgs 8786586, 8787147, 8787153, 8787164, and 8822568 and 1 set of parts for firt article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests be conducted in order specified in Table I.

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Table I

First Article Inspection and Tests

Examination or Test	Dest	Non Dest	Parts	Sub Assy	Assy
1. Classification of characteristics (see 4.3.2.1)		X	X	X	X
2. Rotating band and obturating band (see 3.5)	X			X	
3. Cup spring distortion (see 3.6)	X		X		
4. Seating of rubber ring (see 3.7)	X			X	
5. Air pressure test (see 3.8)		X			X
6. Corrosion resistance (see 4.3.3.2)	X		X		
7. Salt spray (see 4.3.3.3)	X		X		
8. Tape test (see 4.3.3.7)		X			X

4.2.3 Inspection equipment The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with MIL-STD-1168. Changes in lot numbers shall be recorded on Ammunition Data Cards in accordance with MIL-STD-1167 and MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	103	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2)
Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be as follows:

4.3.2.1.1 Nose plug, prior to painting (see dwg 8787101).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Perpendicularity of underside of lip with pitch diameter of threads.....Gage

Minor:

201. Total length.....Gage
202. Pitch diameter of threads.....Gage
203. Major diameter of threads, min.....Gage
204. Diameter of cavity, max.....Gage
205. Depth of cavity, max.....Gage
206. Diameter of thread undercut below lip, min.....Gage
207. Length from nose end to lip.....Gage
208. Thickness of lip adjacent to wrench slots, min.....Gage
209. Width of wrench slots, min (functional).....Gage
210. Distance between wrench slots.....Gage
211. Basic taper per inch correct.....Gage
212. Finish proper.....Visual
213. Stamping correct, complete and identifiable.....Visual
214. Workmanship (see 3.10).....Visual/Manual

4.3.2.1.2 Pin shoulder headless (see dwg 8804300).

Categories	Characteristics	Method of Inspection

Critical: None defined.

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Major:

101. Diameter of shaft.....Gage
102. Concentricity of shaft diameter with pitch diameter of external thread.....Gage

Minor:

201. Pitch diameter of external thread.....Gage
202. Major diameter of external thread.....Gage
203. Minor diameter of internal thread.....Gage
204. Pitch diameter of internal thread.....Gage
205. Depth of threaded hole, max.....Gage
206. Length of internal thread, min.....Gage
207. Width of slot, min.....Gage
208. Depth of slot, min.....Gage
209. Total length.....Gage
210. Length of shaft, min.....Gage
211. Finish proper.....Visual
212. Workmanship (see 3.10).....Visual/Manual

4.3.2.1.3 Ogive (see dwg 8787102).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Concentricity of pitch diameter of internal threads with pitch diameter of external threads including perpendicularity to front and rear shoulders.....Gage
102. Pitch diameter of internal threads, max.....Gage
103. Minor diameter of internal threads, max.....Gage
104. Pitch diameter of external threads, min.....Gage
105. Major diameter of external threads, min.....Gage
106. Perpendicularity of rear shoulder with external thread.....Gage

Minor:

201. Pitch diameter of internal threads, min.....Gage
202. Pitch diameter of external threads, max.....Gage
203. Major diameter of external threads, max.....Gage
204. Concentricity of point at forward external location with pitch diameter of external threads.....Gage

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205. Total length.....Gage
206. Length from rear end to flange.....Gage
207. Thickness of lip adjacent to wrench slots, min.....Gage
208. Width of wrench slots, max.....Gage
209. Distance between wrench slots.....Gage
210. Inside diameter at rear.....Gage
211. Concentricity of inside diameter at rear with pitch diameter of external threads.....Gage
212. Thickness from front end to shoulder of inside cavity.....Gage
213. Length to rear external datum.....Gage
214. Length to forward external datum.....Gage
215. Depth to rear internal datum.....Gage
216. Depth to forward external datum.....Gage
217. Diameter of external thread undercut.....Gage
218. Width of external thread undercut, max.....Gage
219. Stamping present, correct, not mislocated or illegible.....Visual
220. Protective finish proper (no scratches or bare spots).....Visual
221. Workmanship (see 3.10).....Visual/Manual

4.3.2.1.4 Screw machine (see dwg 8804302)

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:	None defined.	
Minor:		
201.	Pitch diameter of thread	Gage
202.	Major diameter of thread	Gage
203.	Length of thread, min	Gage
204.	Total length	Gage
205.	Thickness of flange	Gage
206.	Diameter of flange	Gage
207.	Depth of slot, min	Gage
208.	Width of slot, min	Gage
209.	Protective finish proper	Visual
210.	Workmanship (see 3.10)	Visual/Manual

4.3.2.1.5 Gasket (see dwg 8795639)

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:

101. No split, cut, scrap or fold.....Visual/Manual

Minor:

201. Thickness.....TMC (see 3.4)
202. Outside diameter, min.....TMC (see 3.4)
203. Inside diameter.....TMC (see 3.4)

4.3.2.1.6 Shield support assembly (see dwg 8786586 covering details of dwg 8786587).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Pitch diameter of internal threads at front end, max.....Gage
102. Minor diameter of internal threads at front end, max.....Gage
103. Pitch diameter of internal threads in squab holes, max.....Gage
104. Minor diameter of internal threads in squab holes, max.....Gage
105. Pitch diameter of internal threads at rear end, max.....Gage
106. Minor diameter of internal threads at rear end, max.....Gage
107. Concentricity of diameter of cavity below internal threads at front end with outside diameter to rear of external lip.....Gage
108. Perpendicularity of floor of cavity below internal threads at front end with sidewall of same cavity.....Gage
109. Perpendicularity of floor of large cavity with diameter of same cavity.....Gage
110. Concentricity of diameter of large cavity with outside diameter.....Gage
111. Location of squab assembly holes.....Gage
112. Concentricity of diameter of center hole in web with outside diameter to rear of external lip.....Gage

Minor:

201. Pitch diameter of internal threads at front end, min.....Gage
202. Pitch diameter of internal threads in squab assembly holes, min.....Gage

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203. Pitch diameter of internal threads at rear end, min.....Gage
204. Depth of large cavity.....Gage
205. Diameter of counterbore at rear end, max.....Gage
206. Depth of counterbore at rear end, max.....Gage
207. Diameter of counterbore at front end, max.....Gage
208. Depth of counterbore at front end, max.....Gage
209. Total length, max.....Gage
210. Location of wrench slots.....Gage
211. Width of wrench slots.....Gage
212. Diameter of hole in center of web.....Gage
213. Length of forward outside diameter.....Gage
214. Finish proper.....Visual
215. Anti-galling coating on internal threads at front end does show a uniform satin texture, but does not show spatter, sags, runs, bare spots, or lumps.....Visual
216. Stamping present, correct, not mislocated, or illegible.....Visual
217. Length of internal thread from rear, min.....Gage
218. Length of internal thread from front end, min....Gage
219. Distance between wrench slots.....Gage
220. Workmanship (see 3.10).....Visual/Manual

4.3.2.1.7 Stop target (see dwg 8787239).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Diameter of large boss.....Gage
102. Diameter of small boss.....Gage
103. Total length.....Gage

Minor:

201. Diameter of flange.....Gage
202. Length from small boss to flange shoulder.....Gage
203. Length from small boss to large boss.....Gage
204. Width of flange slot, min.....Gage
205. Depth of flange slot, min.....Gage
206. Radius at corner between flange and large boss.....Gage
207. Radius at corner between large boss shoulder and small boss outside diameter, max.....Gage
208. Chamfer present at corner between flange outside and flange shoulder.....Visual
209. Finish proper.....Visual

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210. Marking correct, complete, not mislocated or illegible.....Visual
211. Workmanship (see 3.10).....Visual/Manual

4.3.2.1.8 Base plug assembly (see dwg 8822568)

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Gap between shoulder of tube and shoulder of base plug (including specified torque)	Gage
102.	Total length, max.	Gage
103.	Pins present and staked	Visual/Manual
Minor:		
201.	Stamping correct, complete, not mislocated or illegible	Visual
202.	Workmanship (see 3.10)	Visual/Manual

4.3.2.1.9 Bushing machine thread (see dwg 8787240).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Pitch diameter of threads, min	Gage
102.	Major diameter of threads, min	Gage
Minor:		
201.	Pitch diameter of threads, max	Gage
202.	Major diameter of threads, max	Gage
203.	Thickness, min	Gage
204.	Diameter of hole	Gage
205.	Depth of slot, min	Gage
206.	Width of slot, min	Gage
207.	Finish proper	Visual
208.	Protective coating adequate including no bare spot	Visual
209.	Marking correct, complete, not mislocated or illegible	Visual
210.	Workmanship (see 3.10)	Visual/Manual

4.3.2.1.10 Gun tube after application of anti-galling coating on threads (see dwg 8787166).

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Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of external threads, min	Gage
102.	Major diameter of external threads, min	Gage
103.	Pitch diameter of internal threads at front end	Gage
104.	Minor diameter of internal threads at front end, max	Gage
105.	Pitch diameter of internal threads near center of tube	Gage
106.	Minor diameter of internal threads near center of tube, max	Gage
107.	Pitch diameter of internal threads at rear end, max	Gage
108.	Minor diameter of internal threads at rear end, max	Gage
109.	Outside diameter	Gage
110.	Concentricity of outside diameter with major diameter of external threads including perpendicularity to rear surface	Gage
111.	Concentricity of internal threads at front end with major diameter of external threads including perpendicularity with bottom of cavity	Gage
112.	Inside diameter	Gage
113.	Concentricity of inside long diameter and inside diameter between rear threads with major diameter of internal threads near center of tube	Gage
114.	Inside diameter of tube between rear threads, min	Gage
115.	Diameter of cavity below internal threads at front end, min	Gage
116.	Finish on outside diameter proper	Visual
117.	Chamfer on external threads present	Visual
Minor:		
201.	Pitch diameter of external threads, max	Gage
202.	Pitch diameter of internal threads at rear, min	Gage
203.	Diameter of cavity below internal threads at front end, max	Gage
204.	Depth to bottom of cavity below internal threads at front end	Gage
205.	Diameter of counterbore at front end	Gage

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206. Depth of counterbore at front end, max.....Gage
207. Width of undercut below internal threads at front end, max.....Gage
208. Diameter of undercut forward of internal threads near center of tube.....Gage
209. Inside diameter toward rear, max.....Gage
210. Diameter of undercut forward of internal threads at rear.....Gage
211. Width of undercut forward of internal threads toward rear, max.....Gage
212. Width of undercut forward of internal threads at center of tube, max.....Gage
213. Total length.....Gage
214. Diameter of external thread lead.....Gage
215. Depth of external thread lead, max.....Gage
216. Location of external threads toward rear.....Gage
217. Radius between bottom and sidewall of cavity below internal threads at front end correct, max.....Gage
218. Radius or chamfer between bottom of cavity below internal threads at front end and inside diameter present.....Visual
219. Finish proper (exclusive of outside diameter).....Visual
220. Feather edge removed from external threads as specified.....Visual
221. Location from rear of internal threads near center.....Gage
222. Finish of anti-galling coating on external thread, outside diameter, and forward internal thread does show uniform satin texture but does not show spatter, sags, runs, bare spots, or lumps.....Visual
223. Workmanship (see 3.10).....Visual/Manual
224. Stamping correct, complete, not mislocated, or illegible.....Visual

4.3.2.1.11 Cone disc after plating (see dwg 8787154).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Thickness, min.....Gage
102. Location of each hole in the three hole sets (nine holes) with outside diameter.....Gage
103. Location of two locating pin holes with outside diameter.....Gage

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Minor:

201. Diameter of each hole in the three hole sets (nine holes).....Gage
202. Outside diameter.....Gage
203. Diameter of locating pin holes (two holes).....Gage
204. Radius or chamfer on outside diameters present (2 places).....Visual
205. Finish proper.....Visual
206. Stamping correct, complete, not mislocated or illegible.....Visual

4.3.2.1.12 Body section rear prior to banding and application of anti-galling coating on applicable surface (see dwg 8787164).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major: None defined.

Minor:

201. Location of rotating band seat from base end.....Gage
202. Width of rotating band seat.....Gage
203. Diameter of rotating band seat.....Gage
204. Height of waves on rotating band seat.....Gage
205. Flat at top front of rotating band seat, max.....Gage
206. Radius present at top rear of rotating band seat, max.....Visual
207. Ribs do wave as they circumvent band seat.....Visual
208. Chisel cut number correct in band seat ribs and properly spaced to extremes.....Visual

4.3.2.1.13 Body section rear assembly including application of anti-galling coating on applicable surfaces (see dwg 8787164).

Categories	Characteristics	Method of Inspection

Critical:

1. Length from base end to internal datum diameter on sharp taper forward of threads.....Gage

Major:

101. Diameter of projection to rear of large groove in band.....Gage

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102. Diameter of large rotating band projection.....Gage
103. Concentricity of large rotating band projection with outside diameter of body.....Gage
104. Diameter of flats forward of large rotating band projectile.....Gage
105. Concentricity of flats forward of large rotating band projection with outside diameter of body.....Gage
106. Rotating band undercut not excessive, max. (where applicable).....Gage
107. Pitch diameter of internal threads base end, max.....Gage
108. Minor diameter of internal threads at base end, max.....Gage
109. Pitch diameter of internal threads in creep stop cavity, max.....Gage
110. Minor diameter of internal threads in creep stop cavity, max.....Gage
111. Pitch diameter of external thread, min.....Gage
112. Major diameter of external thread, min.....Gage
113. Perpendicularity of rear face surface with outside diameter.....Gage
114. Concentricity of major diameter of internal thread at base end with outside diameter including perpendicularity with rear surface.....Gage
115. Concentricity of outside diameter with external thread.....Gage
116. Concentricity of inside diameter between large internal thread cavity and front large counterbore with major diameter of internal thread at base end including perpendicularity with base end.....Gage
117. Perpendicularity of bottom surface dashpot cavity diameter with base end surface.....Gage
118. Perpendicularity of bottom surface dashpot cavity diameter with base end surface.....Gage
119. Outside diameter of body.....Gage
120. Diameter of creep stop holes.....Gage
121. Depth of creep stop thread cavity.....Gage
122. Location of dashpot holes.....Gage
123. Diameter of dashpot holes.....Gage
124. Location of small hole in bottom shoulder of dashpot cavity.....Gage
125. Thickness from bottom of inside dashpot cavity to outside cylinder surface.....Gage
126. Inside diameter between large internal thread cavity and front large counterbore.....Gage
127. Length from base end to undercut bottom shoulder.....Gage

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128. Finish proper on inside surface as specified.....Visual
129. Diameter of small hole in bottom shoulder of dashpot cavity.....Gage
130. Major diameter of internal thread at base end, max.....Gage

Minor:

201. Length from projection at rear of large groove to end of rotating band.....Gage
202. Angle at large groove on rotating band.....Gage
203. Width of flat on rotating band projection,min...Gage
204. Length from rotating band projection flat to end of rotating band.....Gage
205. Length from rotating band projection to first small groove.....Gage
206. Angle forward of rotating band projection correct.....Gage
207. Width of four small grooves on rotating band.....Gage
208. Width between small grooves on rotating band.....Gage
209. Depth of four small grooves on rotating band.....Gage
210. Width of flat forward small groove on rotating band.....Gage
211. Total width of rotating band.....Gage
212. Pitch diameter of internal threads base end, min.....Gage
213. Pitch diameter of internal threads in creep stop cavity, min.....Gage
214. Pitch diameter of external threads, max.....Gage
215. Location of base end datum diameter.....Gage
216. Radius at mouth of base end taper.....Gage
217. Total length.....Gage
218. Location of creep stop holes.....Gage
219. Depth of threads in creep stop cavities, min.....Gage
220. Length from base end to undercut below internal base end thread.....Gage
221. Diameter of undercut above base end internal threads.....Gage
222. Diameter of dashpot counterbore above hole.....Gage
223. Diameter of dashpot counterbore below hole.....Gage
224. Depth of dashpot counterbore above hole.....Gage
225. Location of wrench slots rear front face surface.....Gage

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226.	Width of wrench slots near front face surface	Gage
227.	Distance between wrench slots	Gage
228.	Diameter of undercut at rear of external threads	Gage
229.	Width of undercut at rear of external threads, max	Gage
230.	Length from front end surface to shoulder below external thread undercut	Gage
231.	Distance from base of body to 6.5000 datum dia	Gage
232.	Radius at corner of dashpot hole and counter-bore below dashpot hole present	Visual
233.	Radius at bottom of wrench slot near front face surface present	Visual
234.	Radius at undercut to rear of external threads present	Visual
235.	Finish proper (outside surface only)	Visual
236.	Feather edge removed from specified threads for a distance of 180°	Visual
237.	Finish of anti-galling coating on external threads, creep stop cavity threads and outside diameter of body does show uniform satin texture, but does not show spatter, sags, runs, bare spots, or lumps	Visual
238.	Angle forward of flat forward of foremost small groove on rotating band present	Visual
239.	Flat present on projection to rear of large groove on rotating band	Visual

4.3.2.1.14 Nut plain round (see dwg 8787113).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Pitch diameter of thread, max	Gage
102.	Minor diameter of thread, max	Gage
Minor:		
201.	Pitch diameter of threads, min	Gage
202.	Outside diameter, min	Gage
203.	Thickness, min	Gage
204.	Depth of slots, min	Gage
205.	Width of slots, min	Gage
206.	No slot missing or incorrect position	Visual
207.	No chamfer missing on specified external corner	Visual

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208. Stamping correct, complete, not mislocated or illegible.....Visual
209. Finish proper.....Visual
210. Workmanship (see 3.10).....Visual/Manual

4.3.2.1.15 Case support disc after application of anti-galling coating on threads (see dwg 8787141).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Pitch diameter of threads.....Gage
102. Major diameter of threads, min.....Gage
103. Outside diameter forward of threads, max.....Gage

Minor:

201. Outside diameter forward of threads, min.....Gage
202. Length of outside diameter forward of threads....Gage
203. Total length.....Gage
204. Diameter of holes.....Gage
205. Location of holes.....Gage
206. Chamfer at forward end proper.....Visual
207. Finish proper.....Visual
208. Workmanship (see 3.10).....Visual/Manual
209. Finish of anti-galling coating on threads does show uniform satin texture, but does not show spatter, sags, runs, bare spots or lumps.....Visual
210. Stamping correct, complete, not mislocated or illegible.....Visual

4.3.2.1.16 Base jacket assembly prior to installing ring and rubber obturating ring (see dwg 8794529).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Perpendicularity of inside shoulder below extensions with diameter of hole.....Gage
102. Concentricity of obturating band seating surface with diameter of hole.....Gage
103. Blending of outside radius on extensions with width of legs proper.....Visual

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104. Tool markings on flat surfaces between extensions proper.....Visual

Minor:

201. Diameter of obturating band seating surface.....Gage
202. Width at top of obturating band groove.....Gage
203. Length from front end surface to obturating band groove.....Gage
204. Radius at bottom corners of obturating band groove, max. (to be controlled by tool manufacture).....Gage
205. Flat at upper edges of obturating band groove, max.....Gage
206. Angle of undercut on obturating band groove correct (to be controlled by tool manufacture).....Gage
207. Outside diameter.....Gage
208. Width of obturating band flange section.....Gage
209. Workmanship (see 3.10).....Visual/Manual

4.3.2.1.17 Base jacket assembly (see dwg 8787153).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Large diameter of ring cavity.....Gage
102. Small diameter of ring cavity.....Gage
103. Concentricity of ring cavity large diameter.....Gage
104. Concentricity of ring cavity small diameter.....Gage
105. Finish proper on ring cavity surface.....Gage
106. Location of small holes.....Gage

Minor:

201. Depth of ring cavity.....Gage
202. Small drilled hole present (2 required).....Visual

4.3.2.1.18 Base jacket and obturating ring assembly (see dwg 8787147).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major: None defined.

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Minor:

201. Outside diameter of extensions.....Gage
202. Width across extensions.....Gage
203. Radius on inside of extensions proper.....Gage
204. Location of extensions.....Gage
205. Thickness of inside diameter (to be controlled by tool manufacture).....Gage
206. Radius present on outside surface of extensions.....Visual
207. Feather edge removed on inside corners of extensions.....Visual
208. Finish proper including front end and center hole.....Visual
209. Marking correct, complete, and identifiable.....Visual
210. Workmanship (see 3.10).....Visual/Manual

4.3.2.1.19 Base jacket and rubber ring assembly (see dwg 8787147 covering a detail of dwg 8787153).

Categories	Characteristics	Method of Inspection

Critical:

1. Inside diameter of rubber ring, min.....Gage
2. No split, crack or gouge.....Visual

Major:

101. Thickness between rubber ring and forward end surface.....Gage

Minor:

201. Workmanship (see 3.10).....Visual/Manual

4.3.2.1.20 Spring helical after seasoning, plating, and treatment to remove embrittlement (see dwg 8787145).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major: None defined.

Minor:

201. Load at assembled minimum height.....Test
202. Load at maximum working height.....Test
203. Solid height, max.....Gage
204. Outside diameter, max.....Gage
205. Inside diameter, min.....Gage

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206. Both spring ends squared and ground.....Visual
207. No bare spot or scratches exposing base metal.....Visual
208. Workmanship (see 3.10).....Visual/Manual

4.3.2.1.21 Cup (see dwg 8787146).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major: None defined.		
Minor:		
201.	Total length, min	Gage
202.	Thickness of bottom, min	Gage
203.	Thickness of sidewall, min	Gage
204.	Inside diameter	Gage
205.	Radius on outside bottom present	Visual
206.	Radius on inside bottom present	Visual
207.	Finish proper	Visual
208.	Workmanship (see 3.10)	Visual/Manual

4.3.2.1.22 Base O-Ring (see dwg 8787167).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major: None defined.		
Minor:		
201.	Inside diameter	TMC(see 3.4)
202.	Diameter of ring	TMC(see 3.4)
203.	No surface defects, cut, tear, voids, etc	Visual
204.	Workmanship (see 3.10)	Visual/Manual

4.3.2.1.23 Base after application of anti-galling coating on threads, but prior to painting (see dwg 8787142).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Pitch diameter of external threads, min	Gage
102.	Major diameter of external threads, min	Gage
103.	Pitch diameter of large internal thread, max	Gage
104.	Minor diameter of large internal threads, max	Gage

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105. Major diameter of large internal threads, max....Gage
106. Pitch diameter of small internal threads, max....Gage
107. Minor diameter of small internal thread, max.....Gage
108. Major diameter of small internal thread, max.....Gage
109. Perpendicularity of extreme rear face with major diameter of small internal thread.....Gage
110. Concentricity of large internal thread at forward and rear locations from front face surface with external thread including perpendicularity with bottom surface of large internal thread cavity.....Gage
111. Perpendicularity of flange front shoulder surface with major diameter of external threads.....Gage
112. Concentricity of outside diameter of flange with major diameter of external threads.....Gage
113. Location of datum diameter at rear end of internal thread.....Gage
114. Chamfer of male thread present.....Visual

Minor:

201. Pitch diameter of external thread, max.....Gage
202. Pitch diameter of large internal threads, min....Gage
203. Total length.....Gage
204. Diameter of external thread undercut.....Gage
205. Distance from front face surface to flange shoulder.....Gage
206. Width of external thread undercut, max.....Gage
207. Diameter of four holes in side of flange.....Gage
208. Location of four holes in side of flange.....Gage
209. Depth of four holes in side of flange, min.....Gage
210. Height of projection at base end.....Gage
211. Radius on projection at base end.....Gage
212. Outside diameter of projection at base end.....Gage
213. Outside diameter at flange.....Gage
214. Depth to bottom of small internal thread cavity.....Gage
215. Width of undercut below small internal thread, max.....Gage
216. Diameter of undercut below small internal thread.....Gage
217. Depth of large cavities in front face.....Gage
218. Width of large cavities in front end.....Gage
219. Distance across directly opposed large cavities in front end.....Gage
220. Depth to bottom of undercut below large internal thread.....Gage
221. Width of undercut below large internal thread.....Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:03Z
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222. Depth of large internal thread counterbore.....Gage
223. Diameter of large internal thread counterbore....Gage
224. Diameter of four holes in front end, min.....Gage
225. Depth of four holes in front end.....Gage
226. Location of four holes in front end.....Gage
227. Diameter of center through hole.....Gage
228. Radius in undercut to rear of external threads correct.....Gage
229. Radius at bottom of large cavities in front end correct.....Gage
230. Chamfer at rear edge of flange diameter present.....Visual
231. Finish proper.....Visual
232. Radius in sidewall of large cavities in front end present.....Visual
233. Feather edge removed from all applicable threads for a distance of 180°.....Visual
234. Finish of anti-galling coating on threads does show uniform satin texture but does not show spatter, sags, runs, bare spots, or lumps.....Visual
235. Workmanship (see 3.10).....Visual/Manual
236. Marking correct, complete, not mislocated or illegible.....Visual
237. Metal not excessively raised due to stamping.....Visual
238. Taper at mouth of small internal cavity present.....Visual

4.3.2.1.24 Base plug after application of anti-galling coating on threads (see dwg 8787144).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Pitch diameter of threads, min.....Gage
102. Major diameter of threads, min.....Gage
103. Perpendicularity of under surface of flange with major diameter of threads.....Gage

Minor:

201. Pitch diameter of threads, max.....Gage
202. Diameter forward of thread, max.....Gage
203. Width of diameter forward of thread, max.....Gage
204. Width of thread undercut, max.....Gage
205. Thickness of flange.....Gage

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206. Total length.....Gage
207. Diameter of flange.....Gage
208. Location of wrench holes.....Gage
209. Depth of wrench holes, min.....Gage
210. Diameter of wrench holes, min.....Gage
211. Radius at O-ring groove, max.....Gage
212. Chamfer on flange present.....Visual
213. Finish proper.....Visual
214. Feather edge removed from thread for 180° as specified.....Visual
215. Finish of anti-galling coating on threads does show uniform satin texture, but does not show spatter, sags, runs, bare spots, or lumps.....Visual
216. Stamping correct, complete, not mislocated or illegible.....Visual
217. Workmanship (see 3.10).....Visual/Manual

4.3.2.1.25 Spacer, sleeve (see dwg 8794333).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. No split or cracks or separation in excess of 1/2 inch in length.....Visual

Minor:

201. Outside diameter.....Gage
202. Inside diameter.....Gage
203. Total length.....Gage
204. Varnish coating present.....Visual

4.3.2.1.26 Dashpot assembly atomic weapon (see dwg 8794227 and MIL-D-48016); disc solid plain (see dwg 8787112 and MIL-D-50837); ring (see dwg 8787150 and MIL-P-50891); base jacket assembly (see dwg 8787149 and MIL-P-50891); propellant charge nuclear component (see dwg 8787135 and MIL-P-48194); grommet assembly (see dwg 8795680 and MIL-G-48204); plug closing (see dwg 8787108 and MIL-P-50891); plug, machine thread (see dwg 8787104 and MIL-P-50835); cylinder sintered (see dwg 8787107 and MIL-P-50891); body section, atomic projectile, front (see dwg 8787105 and MIL-P-50891); container, shipping and storage, atomic projectile: M500 (see dwg 8795690 and MIL-C-50565) covering details of dwgs 8787149, 8786587, 8787163, 8787105, and 8787244.

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Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) for component in accordance with applicable specification	Visual
Minor: None defined.		

4.3.2.1.27 Shipping assembly prior to assembling base, jacket, ring assembly; body section rear; spacer sleeve; dashpot assemblies; nut plain round; stop target; bushing machine thread; body section front; gasket; ogive and nose plug (see dwg 8786587).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	O-ring adequately assembled to base	Visual/Manual
102.	No cup spring damaged including coil broken or bent out of line	Visual/Manual
Minor: None defined.		

4.3.2.1.28 Shipping assembly prior to assembling, body section rear; spacer sleeve; dashpot assemblies; nut plain round; stop target; bushing machine thread; body section front; gasket; ogive and nose plug (see dwg 8786587).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	No obturating band damaged in any way	Visual/Manual
102.	No rubber obturating ring damaged or not seating properly	Visual/Manual
103.	No gun tube loose	Visual/Manual
Minor: None defined.		

4.3.2.1.29 Shipping assembly prior to assembling, body section front; gasket; ogive and nose plug (see dwg 8787587).

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Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Body section rear fully seated to base plug seating shoulder (including body being tight)	Visual/Manual
102.	Dashpot assemblies tightly assembled	Visual/Manual
103.	Stop target tightly assembled	Visual/Manual
104.	Bushing machine thread adequately staked	Visual/Manual
105.	Spacer adequately positioned between base jacket, and ring assembly and dashpot assemblies	Visual/Manual
Minor: None defined.		

4.3.2.1.30 Shipping assembly (see dwg 8786587).

Categories	Characteristics	Method of Inspection
Critical:		
1.	Weight	Approved Weighing Device
Major:		
101.	Outside diameter of bourrelet ring maximum (max)	Gage
102.	Body section front tightly assembled	Visual/Manual
103.	Ogive threads coated or burnished (prior to assembly)	Visual
104.	Ogive tightly assembled	Visual/Manual
105.	Rubber gasket present between body front and ogive	Visual
106.	Presence of 4 headless shoulder pin; 4 packing, preformed; 4 screw machine	Visual/Manual
107.	Gap between body front and body rear	Gage
108.	Stamping correct, complete, and legible	Visual
109.	Exterior coating not damaged (including no scratches)	Visual
Minor:		
201.	Grommet adequately assembled	Visual/Manual

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202. Workmanship (see 3.10).....Visual/Manual
203. Serialization (see 3.9).....Visual

4.3.2.1.31 Ammunition container assembly prior to assembling any packing materials in ammunition container (see dwg 8787241 sheet 1 of 2).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major: None defined.		
Minor:		
201.	Packing materials thoroughly dry	Visual/Manual
202.	No packing materials excessively damaged to the extent that its use would impair the finished packing assembly	Visual/Manual

4.3.2.1.32 Ammunition container assembly prior to packing projectile, short spacer tube, rear support, filler and cover (see dwg 8787241 sheet 1 of 2).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Vapor barrier envelope with weapons log book present and adequately secured	Visual/Manual
Minor:		
201.	Cushion at bottom present	Visual/Manual
202.	Front support present	Visual/Manual
203.	Long spacer tube present	Visual

4.3.2.1.33 Ammunition container assembly prior to assembling gasket and cover (see dwg 8787241 sheet 1 of 2).

Categories	Characteristics	Method of Inspection
Critical: None defined.		

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:03Z
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Major:

101. Desiccant bags properly secured to rear support.....Visual/Manual
102. Humidity indicator properly positioned and secured on cover.....Visual/Manual
103. Hole present in filler and top cushion.....Visual
104. Desiccant holder and cushion present (including desiccant).....Visual/Manual
105. Packing list present (including packing list correct, complete and identifiable).....Visual/Manual

Minor:

201. Filler present.....Visual/Manual
202. Cushion present.....Visual/Manual
203. Short spacer tube present.....Visual

4.3.2.1.34 Ammunition container assembly (see dwg 8787241).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Torque of cover screw correct.....Gage
102. Plug adequately seated.....Visual/Manual
103. Marking correct, complete, and legible.....Visual
104. Inspector's acceptance stamp located correctly.....Visual
105. Plug gasket present.....Visual
106. Car seal present and secure.....Visual/Manual

Minor:

201. Marking located properly.....Visual
202. Exterior coating not damaged (such as dents or scratches).....Visual
203. Serialization (see 3.9).....Visual

4.3.2.1.35 Ammunition container, metal, for accessory parts for 8 Inch, M422 Projectile, assembly, details, packing and marking, prior to assembling cover, top support, accessory parts, and miscellaneous fillers, cushions, and spacers (see dwg 8787244).

Categories	Characteristics	Method of Inspection

Source: https://assist.dla.mil -- Downloaded: 20260410-26T04:03Z
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Critical: None defined.

Major: None defined.

Minor:

201. Any packing materials thoroughly dry.....Visual/Manual
202. Large cushion on bottom of container present.....Visual
203. Large filler above large cushion present.....Visual
204. No bottom support damaged or any component missing or improperly nailed.....Visual
205. Drum liner present.....Visual

4.3.2.1.36 Ammunition container, metal for accessory parts for 8 Inch, M422 Projectile, assembly, details, packing and marking prior to assembling cover, top support and miscellaneous fillers (see dwg 8787244).

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Packing list present (including packing list correct, complete, and identifiable).....Visual/Manual

Minor:

201. Small cushion under plug present.....Visual
202. Plug present.....Visual
203. Small cushion under propellent case assembly present.....Visual
204. Propellent case assembly present.....Visual
205. Small cushion over case support disc present.....Visual
206. Case support disc present.....Visual
207. Sufficient number of small fillers on top of case support disc to insure tight pack.....Visual
208. Desiccant properly positioned and secured to bottom support.....Visual
209. Proper number of desiccant units.....Visual
210. Can containing fuze assembly and igniter present.....Visual
211. Either of two spare O-ring envelopes or two O-rings in each envelope present.....Visual/Manual

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4.3.2.1.37 Ammunition container, metal for accessory parts for Projectile 8 Inch, M422, assembly, details, packing and marking prior to assembling cover (see dwg 8787244).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major: None defined.		
Minor:		
201.	Top support present and properly assembled	Visual
202.	Either of two screw eyes or screw hook present	Visual
203.	Thermometer properly and securely held in position by screw eyes and screw hook	Visual
204.	Wrap containing one steel disc (bottom projectile insert) and also one beryllium disc (top projectile insert) present	Visual
205.	Both of two squab carrying case assemblies present	Visual
206.	Base plug present	Visual
207.	No required component or assembly missing from top support	Visual
208.	No large cushion missing from above top support	Visual
209.	Sufficient number of fillers above top support to insure tight pack	Visual
210.	No humidity indicator missing from area between top support and large cushion on top	Visual

4.3.2.1.38 Ammunition container, metal, for accessory parts for Projectile 8 Inch, M422, assembly, details, packing and marking (see dwg 8787244).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Air vent in bottom of drum present and secure	Visual/Manual
Minor:		
201.	Cover properly positioned and secure	Visual/Manual

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202. Car seal properly attached thru hole in bolt.....Visual
203. No exterior paint damaged as evidenced by scratches or bare spots.....Visual
204. Any required marking present, correct, not mislocated, or illegible.....Visual
205. No dents or other damage.....Visual
206. Workmanship (see 3.10).....Visual/Manual
207. Serialization (see 3.9).....Visual

4.3.3 All tests shall be performed. Tests need not be performed in the order specified.

Test	Test Classification
a. Seating of rotating band and obturating band.	Critical
b. Physical properties of rotating band and obturating band.	Major
c. Corrosion resistance on nose plug, Ogive and cup.	Major
d. Salt spray on plug, machine thread, disc solid plain, cone disc, nut plain round, spring helical and bushing machine thread.	Minor
e. Spring helical distortion.	Major
f. Potting of rubber ring assembly.	Major
g. Air pressure test.	Critical
h. Tape test.	Minor

Of these tests, c, d, and f, are considered destructive and samples so tested shall not be returned to the lot.

4.3.3.1 Rotating band and obturating band.

4.3.3.1.1 Band seating (see 3.5.3). Prior to machining of the rotating bands, bodies or base jackets shall be selected at random from each banding lot as required by Table IV. If the average (each average is based on the three readings at each station) of one set of sample results or the sum of the

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averages of more than one sample is equal to, or less than, the cumulative acceptance number (AN) in Table IV, the banding lot shall be accepted for this characteristic. If the average of one set of sample results or the sum of the averages of more than one sample is equal to or greater than the corresponding cumulative rejection number (RN), the banding lot shall be rejected for this characteristic. If the average of one set of sample results or the sum of the averages of the cumulative sample is greater than the corresponding AN value, but less than the RN value, the result is indecisive and sampling shall continue until decision is reached.

Table IV

Rotating Band and Obturating Band Seating

Rotating Band Stations 1 and 5 and Obturating Band leading and trailing edge (as applicable).

Units Tested (Sample Size)	Cumulative Acceptance Value (AN)	Cumulative Rejection Value (RN)
1	.0046	.0072
2	.0105	.0130
3	.0164	.0189
4	.0223	.0248
5	.0282	.0307
6	.0341	.0366

For acceptance and rejection numbers beyond 6 samples, add .0059 to both columns for each additional sample.

Stations 2, 3, and 4

Number of Units Tested	Cumulative Acceptance Value (AN)	Cumulative Rejection Value (RN)
1	.0107	.0204
2	.0263	.0360
3	.0419	.0516
4	.0575	.0672
5	.0731	.0828
6	.0887	.0984

For acceptance and rejection numbers beyond 6 samples, add .0156 to both columns for each additional sample.

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4.3.3.1.2 Physical properties (see 3.5.2). One sample shall be selected from each band gap lot for the physical properties test. If the sample fails to comply with the applicable requirement, the banding lot shall be rejected.

4.3.3.2 Corrosion resistance on nose plug, Ogive and cup (see applicable drawings). Sample panels for the above testing shall be selected and action taken in accordance with Specification MIL-A-8625.

4.3.3.3 Salt spray, on plug machine thread, disc solid plain, cone disc, nut solid plain, spring helical and bushing machine thread (see applicable drawings). Sample panels for the above testing shall be selected and action taken in accordance with Specification QQ-Z-325 or QQ-P-416, as applicable. Testing shall be as specified in paragraph 4.4.3.

4.3.3.4 Spring helical distortion (see 3.6). Springs shall be selected as given in Table V from each lot for test. If one or more springs fail to comply with the applicable requirements, the lot shall be rejected.

Table V

Lot Size	Sample Size
2-8	1
9-15	2
16-25	3
26-40	5
41-65	8
66-110	13
111-180	22
181-300	36
301 and Over	MIL-STD-105 (AQL 0.25 Percent)

4.3.3.5 Seating of rubber ring assembly (see dwg 8787147). The Government inspector shall select two (2) sample base jacket and rubber ring assemblies from each 8 hour production. One sample assembly shall be selected at the start and the second sample at the end of each 8 hours production, and inspected in accordance with paragraph 4.4.5. If any sample fails to meet the applicable requirements, the 8 hours production shall be rejected.

4.3.3.6 Air pressure test. Each ammunition container in the lot shall be subjected to this test, (see paragraph 4.4.6). If any container fails to comply with the applicable requirements it shall be classified defective and removed from the lot.

4.3.3.7 Tape test. One sample assembly shall be selected from every 20 painted projectile assemblies or portion thereof (as applicable) and tested in accordance with paragraph 4.4.7. If any sample fails to comply with paint adhesion, Section 3, of TT-C-490, the lot of projectiles shall be rejected.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-8786587. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accurancy tolerances of the test equipment used.

4.4.1 Band seating and physical properties. Prior to removal, the unmachined rotating band shall be measured on three diameters 60° apart. At each of the three diameters, five measurements along the width of the band shall be taken, namely at 1/2 inch (station 1 and 5), at 3/4 inch (station 2 and 4), from forward and rear edges of the band and at the center (station 3), totaling fifteen (15) measurements. Prior to removal, the unmachined obturating band shall be measured on six diameters 30° apart. At each of the six diameters, two measurements shall be taken, namely at the leading and trailing edge of the band, making twelve (12) measurements. The location of the points of measurement shall be marked on the band, along with the orientation of the rotating band and body, obturating band and base jacket as applicable. The rotating band or obturating band as applicable shall be removed in a manner that will keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 30 or 24 measurements as applicable) and the diameter of the band seat at these points shall also be measured. To each of the band seat diameters add the sum of the corresponding two band thickness and subtract the results from the corresponding exterior diameter previously determined. The result will indicate the total diametral clearance between the band and its seat. If

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the band is well seated, a negative or interference may be indicated. When irregular band profiles interfere with measurement (prior to removing the band), the manufacturer is authorized to machine the outside of the band in order to obtain an adequate cylindrical surface. The removed band shall be carefully flattened without hammering and test coupons shall be machined and tested in accordance with ASTM-E8-66.

4.4.2 Corrosion resistance on nose plug, ogive and cup (see applicable drawings). The above parts shall be tested as specified in Specification MIL-A-8625.

4.4.3 Salt spray on plug, projectile bottom insert, cone disc, reservoir retainer, sup spring, and creep stop nut (see applicable drawings). The above parts shall be tested as specified in Specification QQ-Z-325 or QQ-P-416, as applicable.

4.4.4 Spring helical distortion test. The springs shall be placed in a fixture which will hold the ends. A pull shall be slowly applied to the spring until what is determined to be the minimum point of permanent distortion is reached without breaking the spring. (Springs so tested for embrittlement shall be scrapped.)

4.4.5 Seating of rubber ring assembly. The rubber ring, from each selected sample of base jacket and rubber ring assemblies, shall be removed to determine proper forming of the rubber ring to the base jacket dove tail groove. Acceptance of the production shall be that the ring completely fills the dove tail groove as determined by visual examination of the rubber ring after complete removal to determine complete form. Inspection samples, after completion of examination, shall be returned to the lot for replacement of the rubber ring.

4.4.6 Air pressure test. Remove vent cap and apply dry air pressure by any method satisfactory to the contracting officer.

4.4.7 Tape test. The sample shall be tested in accordance with paint adhesion, Section 4, TT-C-490.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The projectile shall be packaged in accordance with dwg 8787241 using the specified moisture barrier.

5.1.2 Level A. The miscellaneous parts shall be packaged in accordance with dwg 8787244 using the specified moisture barrier.

5.1.3 Level C. The projectile shall be packaged in accordance with dwg 8787241 except that the moisture barrier may be omitted.

5.1.4 Level C. The miscellaneous parts shall be packaged in accordance with dwg 8787244 except that the moisture barrier may be omitted.

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5.2 Packing.

5.2.1 Level A and C. The projectile shall be packed in accordance with dwg 8787241.

5.2.2 Level A and C. The miscellaneous parts shall be packed in accordance with dwg 8787844.

5.3 Marking. Marking shall be as specified in accordance with dwgs 8787241 and 8787244 as applicable.

5.4 Data cards (see 6.1.2). Data cards shall be prepared in accordance with MIL-STD-1167. The data card shall be classified "Secret Restricted Data". Data cards shall be distributed in the normal manner, however, data cards will not be assembled in any of the packing containers.

6. NOTES

6.1 Ordering data. Procurement documents should specify:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1 and 5.4). Data specified in 4.3.4.1 and 5.4 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-N-I.

6.4 Instructions for first article inspection (see 4.2.1). Parts may be used in subassemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government. Subassemblies may be used in final assemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government.

6.5 Supersession data. This specification supersedes the requirements of Purchase Descriptions PA-PD-910 (Rev 1), dated 31 May 1962 and PA-PD-1077 (Rev 1), dated 20 February 1962.

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Project No. 1110-A014

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